

Minjae Kim

Senior Researcher, Electronics and Telecommunications Research Institute (ETRI)

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Research Areas

- Robotic control and system integration
- Medical, assistive, and rehabilitation robotics
- AI-based prosthetic and gait control
- Neural and myoelectric interfaces
- Haptics and wearable bio-signal sensing

Professional Experience

Electronics and Telecommunications Research Institute (ETRI)

Senior Researcher

Jul 2026 - Present

- Conduct research on robotic systems and assistive technologies.

Neuromeka

Group Leader, Medical Robot Group

Jan 2026 - Jun 2026

Group Leader, HRI Group

Apr 2024 - Jan 2026

- Designed and implemented control architectures for industrial, service, and medical robotic applications.
- Led system integration for surgical robotic systems, humanoid robotic platforms for surgical assistance, a space robotic arm, mobile robots, and service robot platforms.

- Directed cross-functional development for GMP readiness, medical device certification, safety validation, and full-system testing.
- Built digital-twin and integrated control environments using NVIDIA Isaac Sim, robotic manipulators, and multi-finger robot hands.

March Lab, Pohang University of Science and Technology (POSTECH)

Post-Doctoral Fellow

Supervisor: Dr. Keehoon Kim

Jun 2023 - Apr 2024

- Designed and developed a compact EMG sensor module under 20 mm x 20 mm with integrated circuitry, PCB, microcontroller, and communication capabilities.
- Implemented BLE-based real-time communication for bio-signal acquisition and analysis.
- Developed Android and iOS applications for signal monitoring and analysis.

Center for Bionic Medicine, Shirley Ryan AbilityLab

Post-Doctoral Fellow

Jan 2021 - May 2023

Visiting Post-Doctoral Scholar, supported by Korea Institute of Science and Technology (KIST) and Hanyang University

Jan 2020 - Dec 2020

Supervisor: Dr. Levi Hargrove

- Developed deep neural network models to estimate impedance control parameters from prosthetic sensor data, including IMU and load cell measurements.
- Established a real-time prosthetic leg control framework and conducted experiments with disabled participants.
- Built hardware and software interfaces to deploy control models on an Android-based embedded system for real-time prosthetic operation.
- Trained models on gait data from able-bodied subjects for intent recognition and prosthetic control in disabled patients.
- Generated synthetic gait trajectories using generative adversarial networks to create unseen motions beyond the training dataset.
- Designed selective artifact removal methods for sEMG signals recorded during spinal stimulation experiments with stroke patients.

Center for Intelligent and Interactive Robotics, Korea Institute of Science and Technology (KIST)

Post-Doctoral Fellow

Supervisor: Dr. Keehoon Kim

Sep 2018 - Aug 2019

- Developed a microneedle array-based sEMG system by integrating embedded systems and electrodes.
- Developed sEMG signal-based motion recognition methods.
- Proposed model-based compensation for electrode displacement and robust motion recognition independent of electrode properties.
- Generated synthetic sEMG signals using model-based approaches.

Robotics Lab, POSTECH

Ph.D. Student

Supervisor: Dr. Wan Kyun Chung

Mar 2012 - Aug 2018

- Designed and fabricated polymer- and metal-based microneedle array electrodes, including high-density configurations.
- Evaluated electrode performance under multiple experimental conditions and analyzed parameters affecting signal quality.
- Designed a complete wireless wearable sEMG system pipeline, including electrode design, circuit design, and casing.
- Proposed feature extraction methods for robust motion recognition and validated the model with healthy subjects and an amputee participant.
- Developed a cell stimulation and signal measurement testbed using an XYZ micro-stage for precise cell manipulation.
- Integrated sensors including load cells and thermistors for prosthetic hand evaluation.

Intelligent Mechatronics and Robotics Lab, Chung-Ang University

Summer Intern

Advisor: Dr. Tae-Hyoung Kim

Jul 2011 - Aug 2011

- Studied Constrained Particle Swarm Optimization through literature review.
- Implemented the CPSO algorithm from scratch in MATLAB.

Education

POSTECH

Combined M.S./Ph.D. in Mechanical Engineering

Pohang, Korea

Mar 2012 - Aug 2018

- Supervisor: Dr. Wan Kyun Chung
- Dissertation: *Development of sEMG Interface and Algorithm for Long-term Reliable Measurement*

Chung-Ang University

B.S. in Mechanical Engineering

Seoul, Korea

Mar 2008 - Feb 2012

- Minor in Electrical and Electronics Engineering

Honors and Awards

- Sarah Baskin Outstanding Research Award, 1st place, Shirley Ryan AbilityLab, Jun 2022
- RMI Paper Award, KIST, Dec 2019
- Outstanding Paper Award, International Conference on Ubiquitous Robots, Jun 2018
- Best Paper Award, Korean Conference on Sensors, Nov 2015
- Best Teaser Award, Haptic Symposium, Feb 2014
- Participation Award, Information Search Competition, Chung-Ang University, Nov 2011

Journal Publications

1. **Minjae Kim** and L. J. Hargrove, "A Method for Controlling Prosthetic Legs Using Deep Neural Network Models Trained On Synthetic Gait Trajectory," *Journal of NeuroEngineering and Rehabilitation*, Vol. 20:115, 1-16, Sep 2023.
2. S. Yang, **Minjae Kim**, S. K. Hong, S. Kim, W. K. Chung, G. Lim, and H. Jeon, "Design of 3D Controller Using Nanocracking Structure-Based Stretchable Strain Sensor," *Sensors*, Vol. 23, 1-

14, May 2023.

3. **Minjae Kim** and L. J. Hargrove, "A Gait Phase Prediction Model Trained on Benchmark Datasets For Evaluating a Controller For Prosthetic Legs," *Frontiers in Neurorobotics*, Vol. 16, 1-13, Jan 2023.
4. **Minjae Kim**, A. M. Simon, and L. J. Hargrove, "Seamless and intuitive control of a powered prosthetic leg using deep neural network for transfemoral amputees," *Wearable Technologies*, Vol. 3, e24:1-16, Sep 2022.
5. **Minjae Kim** and L. J. Hargrove, "Deep-Learning to Map a Benchmark Dataset of Non-amputee Ambulation for Controlling an Open Source Bionic Leg," *IEEE Robotics and Automation Letters*, Vol. 7, Issue 4, 10597-10604, Jul 2022.
6. **Minjae Kim**, Y. Moon, J. Hunt, K. A. McKenzie, A. Horin, M. McGuire, K. Kim, L. J. Hargrove, and A. Jayaraman, "A Novel Technique to Reject Artifact Components for Surface EMG Signals Recorded During Walking With Transcutaneous Spinal Cord Stimulation: A Pilot Study," *Frontiers in Human Neuroscience*, Vol. 15, Jun 2021.
7. **Minjae Kim**, W. K. Chung, and K. Kim, "Subject-Independent sEMG Pattern Recognition by Using a Muscle Source Activation Model," *IEEE Robotics and Automation Letters*, Vol. 5, Issue 4, 5175-5180, Jul 2020.
8. **Minjae Kim**, K. Kim, and W. K. Chung, "Simple and Fast Compensation of sEMG Interface Rotation for Robust Hand Motion Recognition," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, Vol. 26, Issue 12, 2397-2406, Dec 2018. Selected as the front cover of the 12th issue in the 26th volume.
9. **Minjae Kim** and W. K. Chung, "Spatial sEMG Pattern-Based Finger Motion Estimation in a Small Area Using a Microneedle-Based High-Density Interface," *IEEE Robotics and Automation Letters*, Vol. 3, No. 1, 234-241, Jan 2018.
10. **Minjae Kim**, G. Gu, K. J. Cha, D. S. Kim, and W. K. Chung, "Wireless sEMG system with a microneedle-based high-density electrode array on a flexible substrate," *Sensors*, Vol. 18, No. 1, 92, Dec 2017.
11. **Minjae Kim**, T. Kim, D. S. Kim, and W. K. Chung, "Curved microneedle array-based sEMG electrode for robust long-term measurements and high selectivity," *Sensors*, Vol. 15, No. 7, 16265-16280, Jul 2015.

Conference Publications

1. D. G. Park, Y. J. Lee, J. B. Han, **Minjae Kim**, and T. K. Yeu, "A Preliminary Study on Fatigue Estimation for Underwater Robot Operator Using sEMG Signals," *IEEE International Conference on Automation Science and Engineering (CASE)*, New Zealand, Aug 2023.

2. **Minjae Kim**, A. M. Simon, K. Shah, and L. J. Hargrove, "Machine Learning-Based Gait Mode Prediction for Hybrid Knee Prosthesis Control," *45th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, pp. 1-6, Australia, Jul 2023.
3. **Minjae Kim**, W. K. Chung, and K. Kim, "Motion Intensity Extraction Scheme for Simultaneous Recognition of Wrist/Hand Motions," *IEEE International Conference on Robotics and Automation (ICRA)*, pp. 10112-10117, Virtual, May 2020.
4. **Minjae Kim**, W. K. Chung, and K. Kim, "Preliminary Study of Virtual sEMG Signal-Assisted Classification," *International Conference on Rehabilitation Robotics (ICORR)*, pp. 1133-1138, Canada, Jun 2019.
5. **Minjae Kim** and W. K. Chung, "Muscle Activation Source Model-based sEMG Signal Decomposition and Recognition of Interface Rotation," *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp. 2780-2786, Spain, Oct 2018.
6. **Minjae Kim**, G. Gu, and W. K. Chung, "Pneumatic Microneedle-Based High-Density sEMG Sleeve for Stable and Comfortable Skin Contact during Dynamic Motion," *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp. 5477-5482, Spain, Oct 2018.
7. **Minjae Kim**, G. Gu, and W. K. Chung, "Pneumatic Sleeve-Assisted Stable sEMG Measurement for Microneedle Array Electrode," *International Conference on Ubiquitous Robots (UR)*, pp. 1-4, USA, Jun 2018.
8. **Minjae Kim**, G. Gu, and W. K. Chung, "Skin Grasping sEMG Interface Based on Microneedle Array Electrode," *International Conference on Ubiquitous Robots and Ambient Intelligence (URAI)*, pp. 839-840, Korea, Jun 2017.
9. **Minjae Kim**, D. S. Kim, and W. K. Chung, "Microneedle-based high-density surface EMG interface with high selectivity for finger movement recognition," *IEEE International Conference on Robotics and Automation (ICRA)*, pp. 973-978, Sweden, May 2016.
10. J. Lee, E. Cho, **Minjae Kim**, Y. Yoon, and S. Choi, "PreventFHP: Detection and Warning System for Forward Head Posture," *Haptic Symposium*, pp. 295-298, USA, Feb 2014.
11. **Minjae Kim**, S. Park, and W. K. Chung, "Flexible Polymer-based Micro Needle Array sEMG Sensor," *International Conference on Ubiquitous Robots and Ambient Intelligence (URAI)*, pp. 1-4, Korea, Oct 2013.

Non-Refereed Publications

1. G. Jeong, Y. Kim, Y. Oh, G. Gu, S. Park, **Minjae Kim**, J. An, and K. Kim, "Design and analysis of underactuated prosthetic hand using slider-crank mechanism," Korea Robotics Society Annual Conference, Jan 2019.
2. G. Gu, **Minjae Kim**, and W. K. Chung, "Development of Wireless High-Density sEMG Acquisition System," Korea Robotics Society Annual Conference, Jan 2018.

3. **Minjae Kim** and W. K. Chung, "sEMG Source Activation Model-Based Motion Estimation Robust to Electrode Placement Change," Korea Robotics Society Annual Conference, Jan 2018.
4. **Minjae Kim** and W. K. Chung, "Novel sEMG-Muscle Model-based Approach for Robust Motion Identification," The Sixth Asia International Symposium on Mechatronics, Sep 2017.
5. G. Gu, **Minjae Kim**, and W. K. Chung, "Design of Portable sEMG Sensor System," Korea Robotics Society Annual Conference, Feb 2017.
6. **Minjae Kim**, T. Kim, D. S. Kim, and W. K. Chung, "Metal Microneedle-based Multi-channel Interface for Biosignal Measurement," Korean Conference on Sensors, Nov 2015.
7. **Minjae Kim**, D. S. Kim, and W. K. Chung, "Reliability of surface electromyographic measurement using microneedle array electrode," International Conference on Biofabrication in Korea, Sep 2014.
8. **Minjae Kim**, T. Kim, D. S. Kim, and W. K. Chung, "Biocompatible microneedle array electrode for surface electromyography," Korean Society of Mechanical Engineers conference of bioengineering, Apr 2014.
9. **Minjae Kim** and W. K. Chung, "ICRA 2013 Review," Journal of Korea Robotics Society, Oct 2013.
10. **Minjae Kim** and W. K. Chung, "The comparison of the sEMG pattern classification performance depends on the electrode impedance," Korea Robotics Society Annual Conference, May 2013.